

Question ID: 2b1a27cd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The quadratic function g models the depth, in meters, below the surface of the water of a seal t minutes after the seal entered the water during a dive. The function estimates that the seal reached its maximum depth of **302.4** meters **6** minutes after it entered the water and then reached the surface of the water **12** minutes after it entered the water. Based on the function, what was the estimated depth, to the nearest meter, of the seal **10** minutes after it entered the water?

Correct Answer: 168

Rationale

The correct answer is **168**. The quadratic function g gives the estimated depth of the seal, $g(t)$, in meters, t minutes after the seal enters the water. It's given that function g estimates that the seal reached its maximum depth of **302.4** meters **6** minutes after it entered the water. Therefore, function g can be expressed in vertex form as $g(t) = a(t - 6)^2 + 302.4$, where a is a constant. Since it's also given that the seal reached the surface of the water after **12** minutes, $g(12) = 0$. Substituting **12** for t and **0** for $g(t)$ in $g(t) = a(t - 6)^2 + 302.4$ yields $0 = a(12 - 6)^2 + 302.4$, or $36a = -302.4$. Dividing both sides of this equation by **36** gives $a = -8.4$. Substituting -8.4 for a in $g(t) = a(t - 6)^2 + 302.4$ gives $g(t) = -8.4(t - 6)^2 + 302.4$. Substituting **10** for t in $g(t)$ gives $g(10) = -8.4(10 - 6)^2 + 302.4$, which is equivalent to $g(10) = -8.4(4)^2 + 302.4$, or $g(10) = 168$. Therefore, the estimated depth, to the nearest meter, of the seal **10** minutes after it entered the water was **168** meters.